

Flocking Modeling, Control, and Optimization of Connected and Automated
Vehicles for Safe and Efficient Mobility

by

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PREVIEW

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ABSTRACT

In large modern urban areas, traffic congestion and fatality have become two serious problems. To improve the safety and efficiency of ground mobility, one promising solution is the cooperative control of connected and automated vehicle (CAV) systems, which can avoid human drivers' incapability and errors. Taking advantage of two-dimensional (2D) vehicular control, this dissertation intends to conduct a thorough investigation of the modeling, control, and optimization of CAV systems with flocking control. Flocking is a dynamic swarm congregating behavior of a group of agents with self-organizing features, and flocking control of CAV systems attempts to achieve the maintenance of a small and nearly constant distance among vehicles, speed match, destination cohesion, and collision and obstacle avoidance.

Concerning artificial multi-agent systems, such as mobile robots and CAV systems, a set of engineering performance requirements should be considered in flocking theory for practical applications. In this dissertation, three novel flocking control protocols are studied, which consider convergence speed, permanent obstacle avoidance, and energy efficiency. Furthermore, considering nonlinear vehicle dynamics, a novel hierarchical flocking control framework is proposed for CAV systems to integrate high-level flocking coordination planning and low-level vehicle dynamics control together. On one hand, using 2D flocking theory, the decision making and motion planning of engaged vehicles are produced in a distributed manner based on shared information. On the other hand, using the proposed framework, many advanced vehicle dynamics control methods and tools are applicable. For instance, in the low-level vehicle dynamics control, in addition to path trajectory tracking, the maintenance of vehicle lateral/yaw stability and rollover propensity mitigation are achieved by using additional actuators, such

as all-wheel driving and four-wheel steering, to enhance vehicle safety and efficiency with over-actuated features.

Co-simulations using MATLAB/Simulink[®] and CarSim[®] are conducted to illustrate the performances of the proposed flocking framework and all controller designs proposed in this dissertation. Moreover, a scaled CAV system is developed, and field experiments are also completed to further demonstrate the feasibility of the proposed flocking framework. Consequently, the proposed flocking framework can successfully complete a 2D vehicular flocking coordination. The novel flocking control protocols are also able to accommodate the practical requirements of artificial multi-agent systems by enhancing convergence speed, saving energy consumption, and avoiding permanent obstacles. In addition, employing the proposed control methods, vehicle stability is guaranteed as expected.

To my mother, Xiaoqin, who loved me

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NOMENCLATURE

α_f	Vehicle front tire sideslip angle
α_r	Vehicle rear tire sideslip angle
β	Vehicle sideslip angle
δ_f	Vehicle front steering angle
δ_r	Vehicle rear steering angle
ϕ	Vehicle roll angle
ψ	Vehicle heading angle
CG	Vehicle center of gravity
F_{cx}	Vehicle corrective longitudinal force
F_{cy}	Vehicle corrective lateral force
F_{re}	Vehicle resistance force
F_{xf}	Vehicle front longitudinal tire force
F_{xr}	Vehicle rear longitudinal tire force
F_{yf}	Vehicle front lateral tire force
F_{yr}	Vehicle rear lateral tire force
I_z	Moment of inertia of vehicle with respect to z-axis
L_f	Distance from the CG to vehicle front wheel center

L_r	Distance from the CG to vehicle rear wheel center
l_s	Vehicle wheel track
M	Vehicle mass
M_z	Vehicle corrective yaw moment
r	Vehicle yaw rate
V	Vehicle global velocity
v_x	Longitudinal vehicle speed
v_y	Lateral vehicle speed
v_f	Vehicle front wheel moving velocity
v_r	Vehicle rear wheel moving velocity

Chapter 1

INTRODUCTION

1.1 Motivation: Why Connected and Automated Vehicle Systems?

In large modern urban areas, traffic congestion and fatality have become two serious problems. Statistically, in 2017, traffic jams resulted in 8.8 billion hours of delayed person-hours and \$179 billion economic loss in the U.S., which is estimated to increase to \$237 billion by 2025 (Schrang, Eisele, and Lomax 2019). Moreover, based on the data obtained from the World Health Organization, traffic accidents caused 1.35 million death worldwide in 2016, whilst it has become the eighth leading cause of death for people of all ages (World Health Organization 2018). In the U.S., the increase in traffic fatalities in 2016, 6%, even reached the highest point in nearly five decades (NHTSA's National Center for Statistics and Analysis 2016). One notable reason for those unexpected problems is that car drivers may not be able to give appropriate commands to avoid traffic congestion or fatality because of human's inattention and incapability to timely observe, predict, and react. It is also reported that 94% of car crashes in the US are due to human errors. Therefore, researchers and engineers propose connected and automated vehicle (CAV) systems as a new promising solution to tackle urban transportation problems by excluding human errors.

Before diving deep into the concept of CAV systems, I would like to define several terminologies used throughout this dissertation.

Autonomous driving technology: the technology drives vehicles only based on perception information and vehicle states acquired from onboard sensors without communication with other vehicles or infrastructures.

Automated vehicle: the vehicle uses mechatronics actuators and autonomous driving technologies to replace human driving.

Connected vehicles: the vehicles are connected together through wireless communication networks to share information.

Using the aforementioned terminologies, the CAV systems in this dissertation refer to a group of automated vehicles that are connected together for cooperative control purposes. Especially, the CAV systems can be explored in the context of vehicular cyber-physical systems (CPS). A general CPS, as a new generation embedded system, integrates a collection of real-time decision making, networking, and physical processing in a feedback loop (Lee 2008; Alur 2015; Lee and Seshia 2016). Specifically, the vehicular CPS is composed of multiple automated vehicles and vehicular wireless communication networks.

The autonomous driving technology is employed as the physical-processing and decision-making layers of vehicular CPS, which enables the autonomy of individual vehicles. On the one hand, the development of vehicle electrification including drive-by-wire technologies paves the way for autonomous driving, which uses electrified actuators to generate control efforts. In other words, the conventional mechanical control systems, such as pedals, steering columns, pumps, and belts, are substituted by electronic control systems. As a result, the drive-by-wire technologies make it possible to drive vehicles using electronic signals, instead of manipulating the mechanical control systems with human intervention. Besides, the drive-by-wire systems also inspire over-actuated vehicle controls, which provides extra control degrees of freedom to maintain

vehicle stability and save energy. On the other hand, powered by the advancement of onboard sensing, data processing, and real-time decision making, various autonomous driving control algorithms for individual vehicles have been prevalently investigated, such as lane-keeping control, pedestrians and obstacles detection, trajectory-tracking control, and adaptive cruise control (ACC) (Martínez-García, Zhang, and Gordon 2016; Sivaraman and Trivedi 2013; Desjardins and Chaib-Draa 2011; Potsaid et al. 2007). A notable success instance of individually autonomous driving was the winner, “Tartan Racing” team, led by Carnegie Mellon University, in the 2007 DARPA Urban Challenge (Darms et al. 2009). Moreover, Waymo LLC. has recently launched to test their fully self-driving cars on public roads in Arizona without a human backup driver, which could be considered as the first Level 4 automated driving car defined by the Society of Automobile Engineers on public roads in the world (Jones 2017).

While the individual automated vehicles are expected to handle most of the complex traffic scenarios, some inherent drawbacks appear. First, the onboard sensors of automated vehicles usually have inherent measurement delays, and it is also time-consuming to observe or estimate states of other vehicles. The time delays in perception not only significantly influence the real-time construction of vehicle surroundings but also jeopardize autonomous driving safety. Second, in some particular situations, the ranging sensors, such as radars, of individual automated vehicles sometimes fail to detect obstacles or other vehicles. For example, a vehicle cannot measure the inter-vehicle gap to its preceding vehicle during cornering as well as the states of the preceding vehicle. Such information absence of other vehicles may result in unexpected safety issues as well. Finally, the cooperative control among a group of automated vehicles is anticipated in intelligent transportation systems to improve safety and traffic capacity, whose objectives include the maintenance of a small and nearly

constant distance among vehicles, speed match, destination cohesion, and collision and obstacle avoidance. However, individual automated vehicles cannot complete those cooperative tasks with the absence of information sharing. Therefore, the vehicular wireless network is required as the communication layer of a vehicular CPS to connect individual automated vehicles together, so-called CAV systems.

Using vehicular wireless communication, the vehicle states and other perception information can be directly acquired without the limitations of sensor delays and state observation/estimation, whilst the perception capability of onboard sensors are also compensated without driving scenario limitations. In addition, the cooperative control can also be achieved based on the shared information and control protocol design. To sum up, the CAV systems have great potential to resolve traffic congestion and fatality issues benefited by vehicle autonomy and vehicular communication.

1.2 Literature Review

1.2.1 Cooperative Control in CAV Systems

Back in the late 1980s, research on the cooperative control of multi-vehicle systems was investigated to achieve consensus phenomena via centralized or distributed control protocols, which referred to dynamical coordination of a group of CAVs connected by wireless communication (Chen and Cheng 2010; Murray 2007; Cao et al. 2012). Benefited from the shared information and coordinated behaviors, the cooperative control of CAV systems has broad engineering applications in transportation systems, military systems, mobile sensor networks, robotic systems, and testbeds (Murray 2007). Recently, with the advancement of vehicular wireless communication, onboard

sensing and data processing, and control-by-wire actuation, the cooperative control of CAV systems is explored in the context of vehicular CPS (Cao et al. 2012).

In the literature, the cooperative control of CAV systems can be considered as a combination of four components, which are agent dynamics, communication topology, distributed controller, and formation geometry, respectively (Li et al. 2015). In addition, three different types of cooperative control of CAV systems can be roughly summarized. The interested readers may refer to (Siegel, Erb, and Sarma 2017; Rios-Torres and Malikopoulos 2016; Guanetti, Kim, and Borrelli 2018) for the detailed survey of CAV systems.

The first type is the cooperative adaptive cruise control (CACC), which is an extension of the ACC (Van Arem, Van Driel, and Visser 2006). The ACC aims to achieve constant distance or time gap among vehicles via automated speed control to release human driver burden (Vahidi and Eskandarian 2003). However, the ACC has issues resulted from inherent sensor delays and string instability. Thus, the CACC typically tries to maintain constant time gaps among multiple vehicles via wireless communication, through which the issues of the ACC are resolved (Milanes et al. 2014). While most of the CACC can control vehicle speeds automatically, the steering commands still need to be provided by human drivers.

The second type of cooperative control of CAV systems is vehicle platooning, which tries to improve transportation capacity, safety, and energy efficiency through a rigid vehicle platoon formation (Jia et al. 2016). Back in the 1980s, many fundamental theories and concepts in the vehicle platooning control were investigated in Partners for Advanced Transportation Technology (PATH) research program at the University of California, Berkeley (Hedrick, Tomizuka, and Varaiya 1994). Compared with the CACC, the platooning can provide additional active steering control and maintain